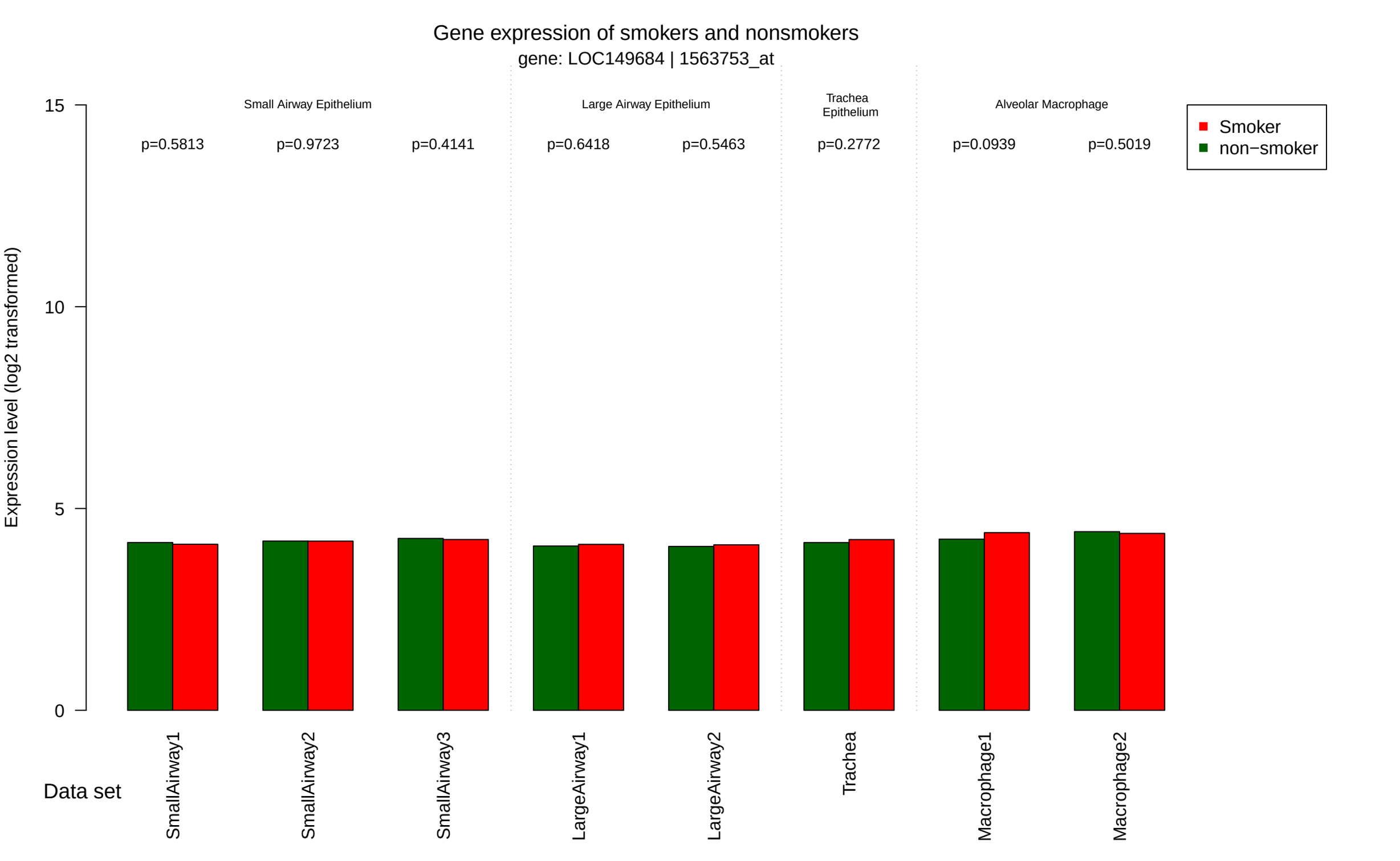
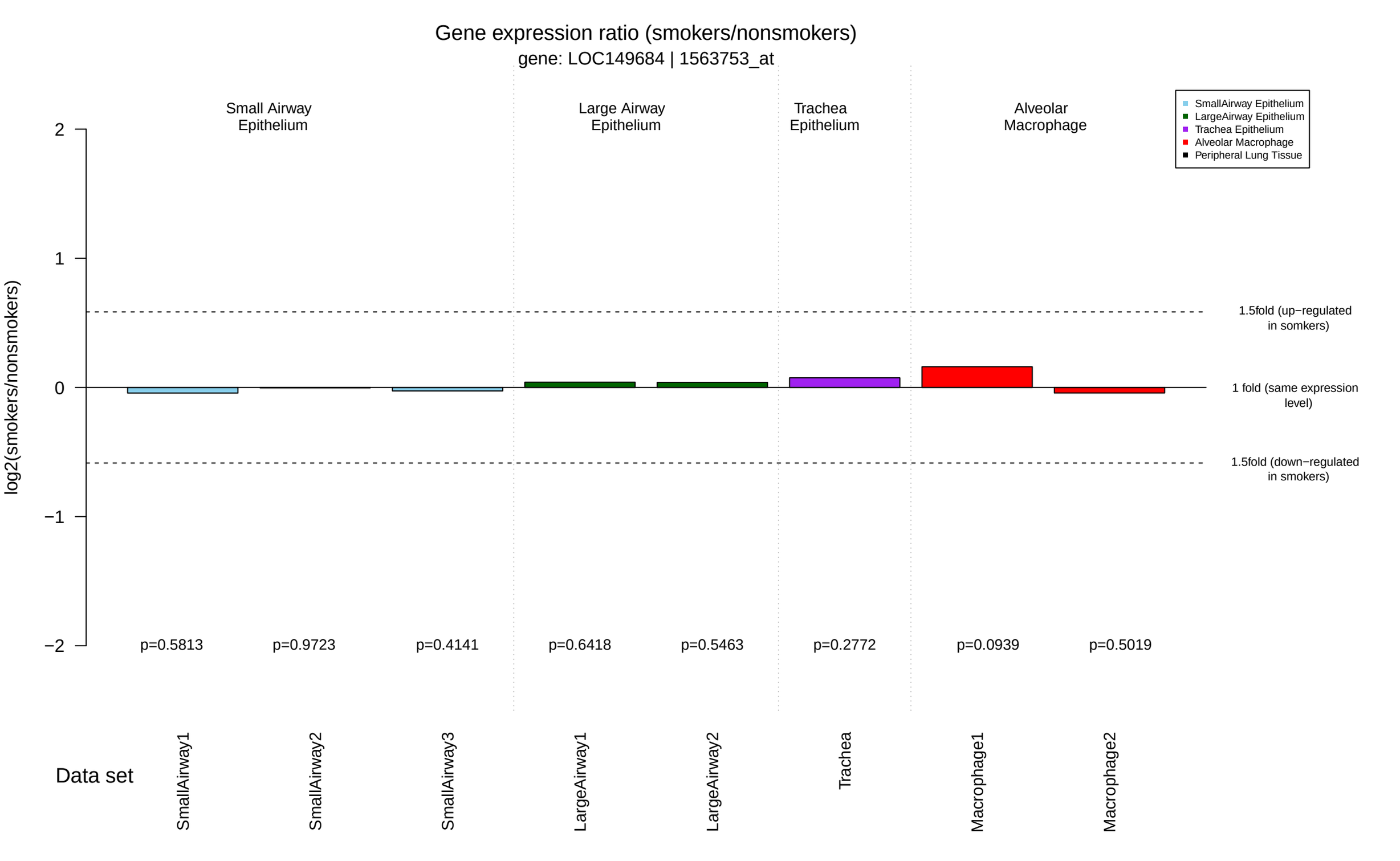
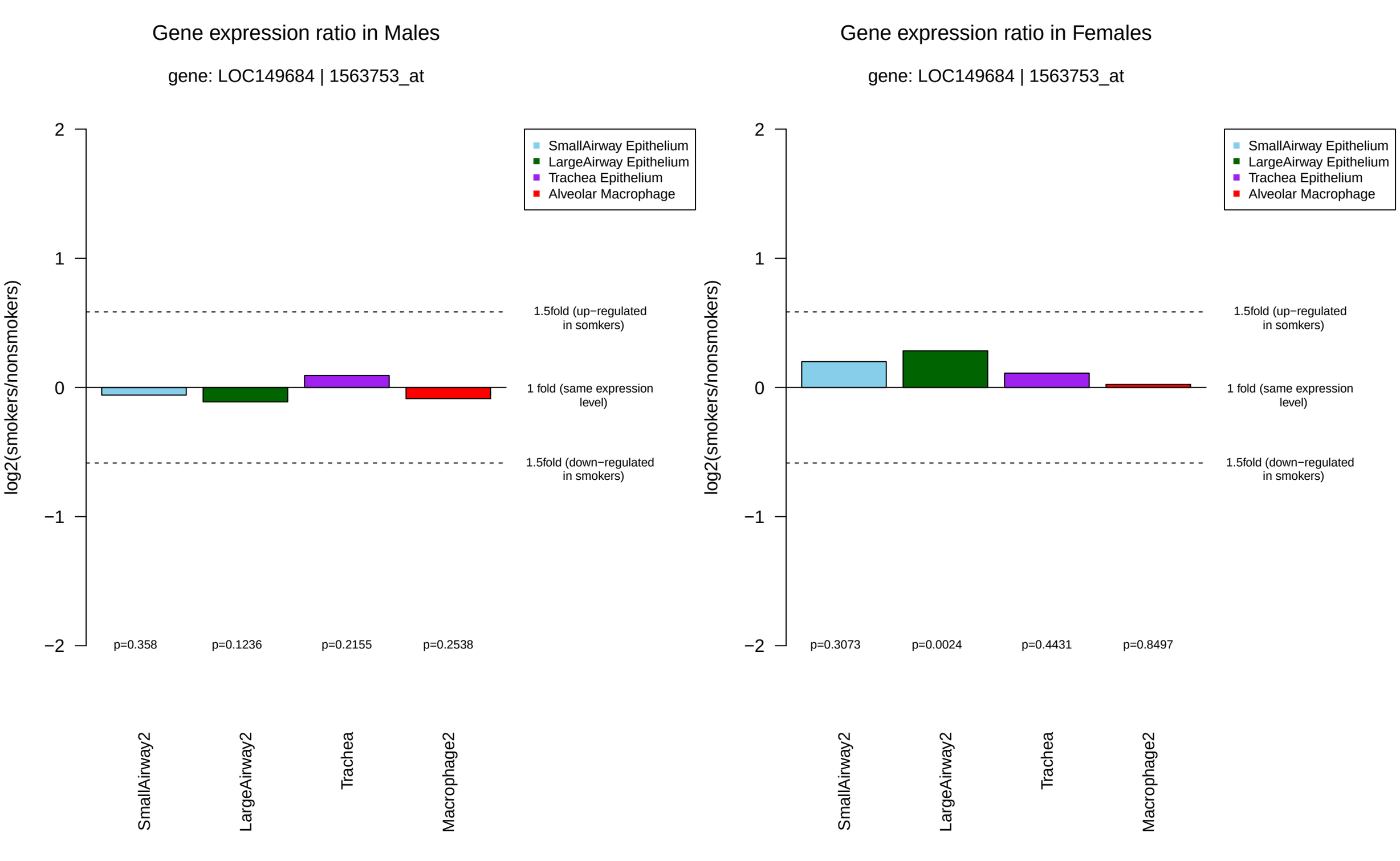
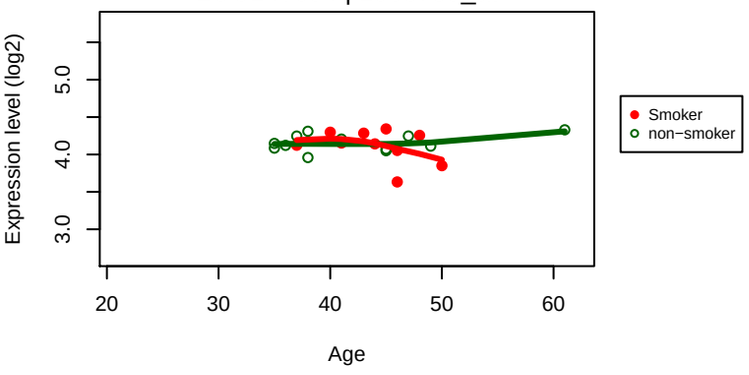
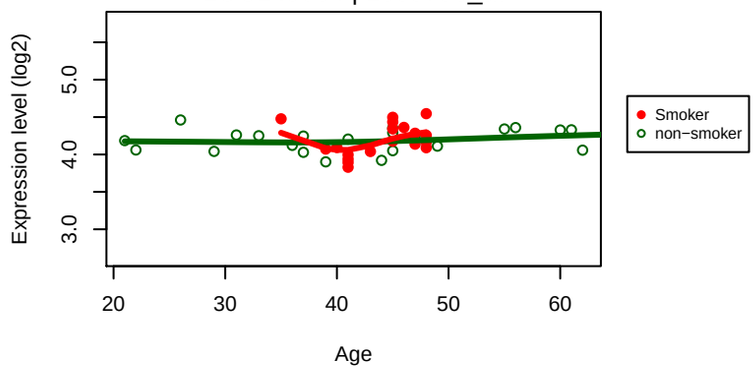
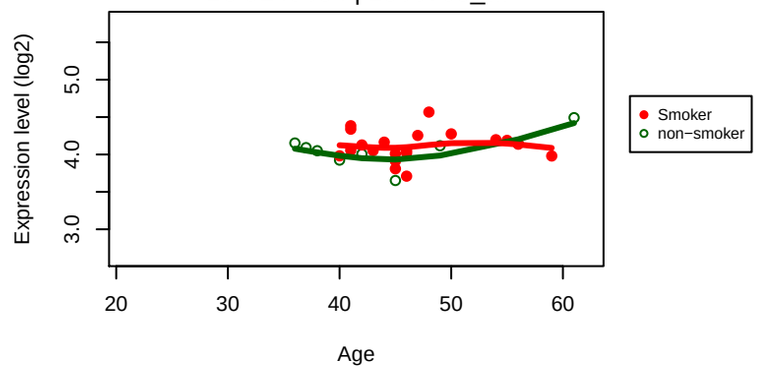

Gene expression of smokers and nonsmokers
gene: LOC149684 | 1563753_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5813
p=0.9723
p=0.4141
p=0.6418
p=0.5463
p=0.2772
p=0.0939
p=0.5019
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

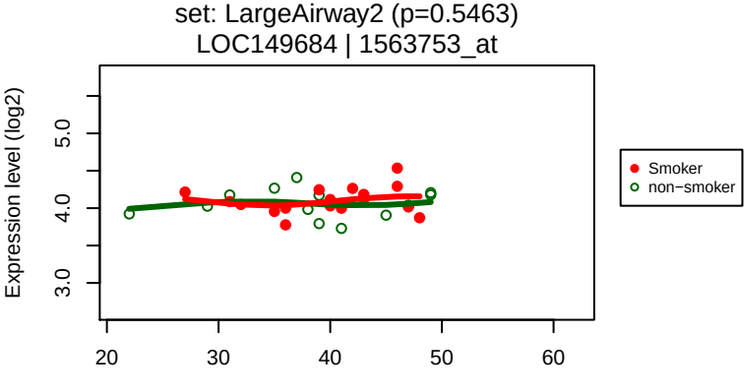

Gene expression ratio (smokers/nonsmokers)
gene: LOC149684 | 1563753_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5813
p=0.9723
p=0.4141
p=0.6418
p=0.5463
p=0.2772
p=0.0939
p=0.5019
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

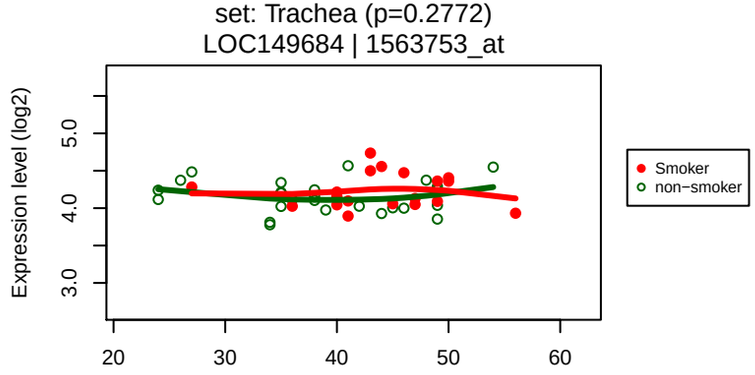

Gene expression ratio in Males
gene: LOC149684 | 1563753_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.358
p=0.1236
p=0.2155
p=0.2538
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC149684 | 1563753_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3073
p=0.0024
p=0.4431
p=0.8497
SmallAirway2
LargeAirway2
Trachea
Macrophage2

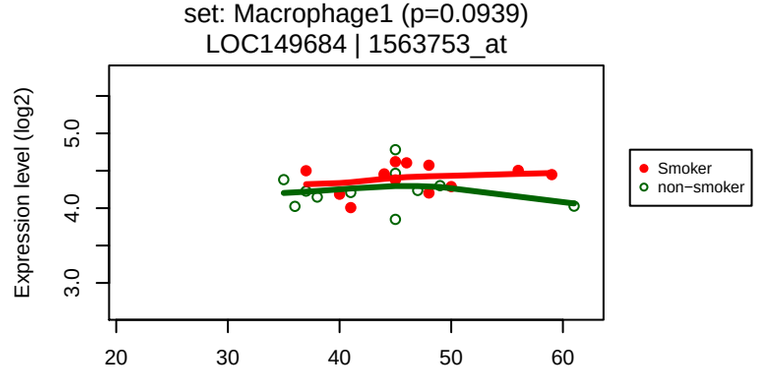

set: SmallAirway1 (p=0.5813)
LOC149684 | 1563753_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker

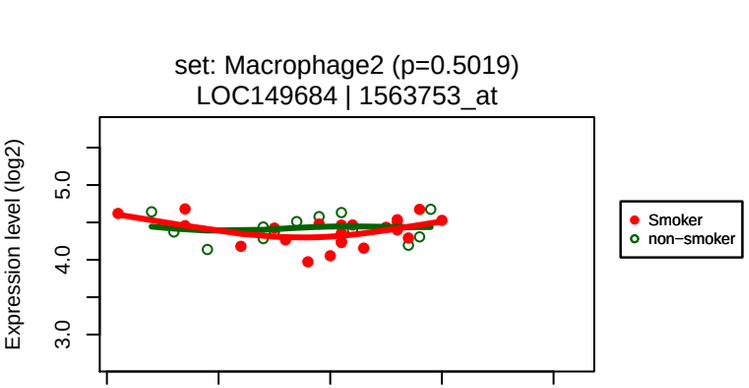

set: SmallAirway2 (p=0.9723)
LOC149684 | 1563753_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker

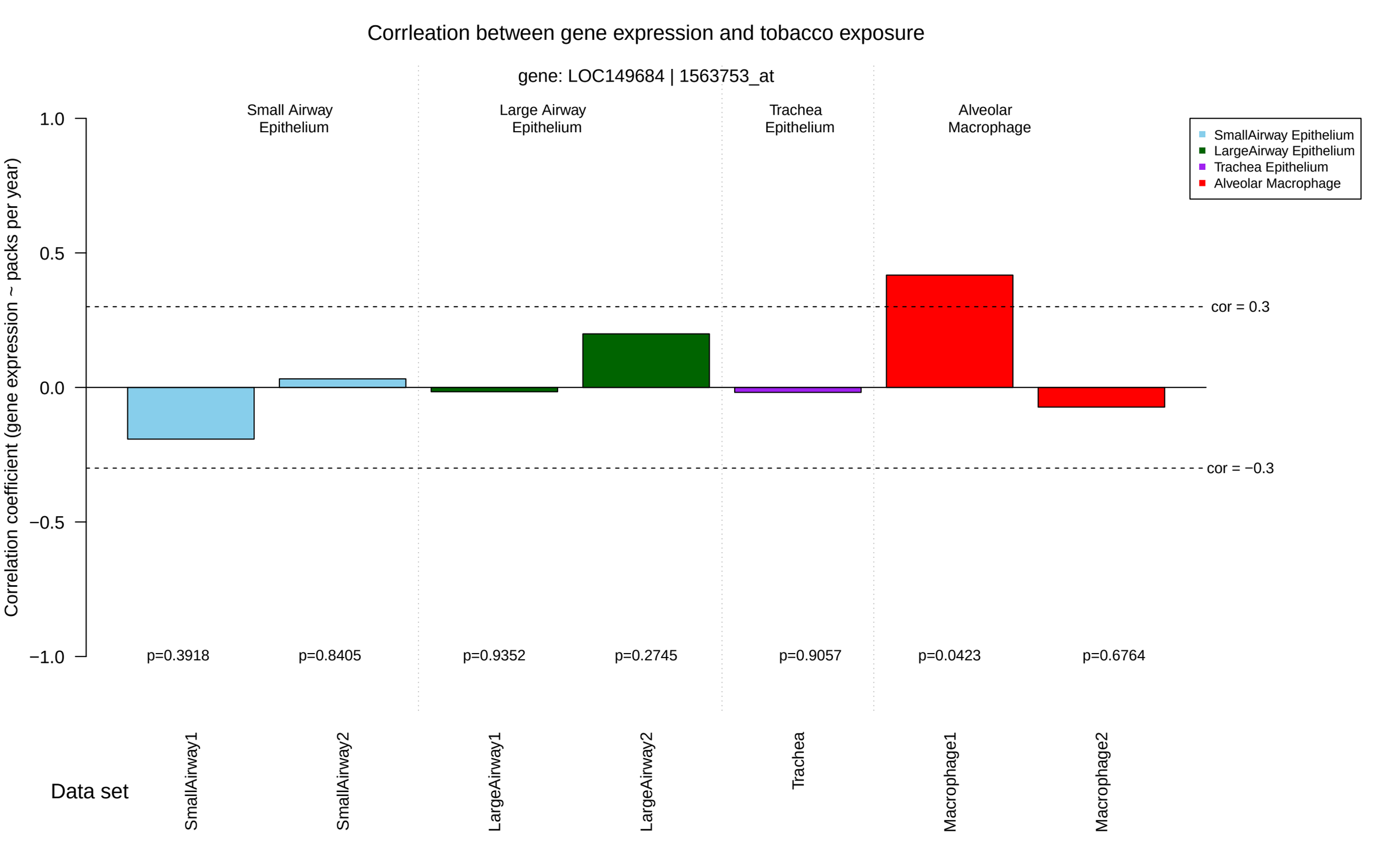

set: LargeAirway1 (p=0.6418)
LOC149684 | 1563753_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5463)
LOC149684 | 1563753_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.2772)
LOC149684 | 1563753_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0939)
LOC149684 | 1563753_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5019)
LOC149684 | 1563753_at
Expression level (log2)
3.0
4.0
5.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC149684 | 1563753_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3918
p=0.8405
p=0.9352
p=0.2745
p=0.9057
p=0.0423
p=0.6764
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2